

Yazzies Catering OMS — Enterprise System Documentation & Audit Report

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1. Executive Business Overview

The Problem

Yazzies Catering faces operational challenges common to Philippine catering SMEs:

- **Double-bookings** — no system to enforce one-event-per-date.
- **Fragmented cash flow** — downpayments tracked on paper with no reconciliation.
- **Staff coordination failure** — assigning cooks/waiters via SMS is unreliable.
- **Inventory blindspot** — no tracking of whether equipment left and returned intact.
- **Zero audit trail** — no record of who changed prices, cancelled bookings, or deleted payments.
- **No online payment path** — manual GCash transfers with no automated confirmation.

The Solution

The OMS digitizes the entire catering lifecycle:

Lifecycle Stage	OMS Module
Lead capture	Client Management + Taste-Test Appointments (schema-ready)
Booking & quoting	Multi-step Booking Wizard with dynamic pricing engine
Payment collection	Cash/GCash/Maya recording + PayMongo online checkout
Staff coordination	Job Order Dispatching with accept/decline workflow
Equipment tracking	Inventory Dispatch & Return with auto-breakage logging
Event execution	Staff Dashboard + Post-Event Report
Financial reporting	Financial Module with revenue analytics
Administration	Superadmin Console + complete Audit Log (445+ entries)

Deployment Strategy: Why This MUST Be Hosted Online

Argument	Technical Detail
PayMongo Webhooks	PayMongo POSTs payment events to a public URL. localhost is unreachable from the internet — online payments can never auto-confirm without a live server.
Client Payment Portal	The invoice link must be accessible to clients on their phones. A localhost URL is meaningless outside the office.

Argument	Technical Detail
Staff Mobile Access	Staff log in from the field (job board, event reports, inventory). Requires a live, public URL.
24/7 Cron Worker	<code>cron_worker.php</code> (email queue, reminders, cleanup) must run on an always-on server — a developer laptop is not a server.
SMTP Deliverability	Gmail/SMTP from localhost is flagged as spam. A proper domain improves email deliverability.
Business Continuity	A single hard-drive failure on the dev machine wipes all booking data. Cloud hosting provides redundancy.
Recommended Stack	Shared hosting (Hostinger/Namecheap), cPanel, PHP 8.1+, MySQL 8.0. Est. cost: ₱2,500–5,000/year.

2. User Roles & System Modules

Account Types & Capabilities

Role: `admin` (Active Administrator)

Single active operator enforced by system. Full access to all modules.

Module	C	R	U	D	Notes
Bookings	✓	✓	✓	✓	Wizard, status changes, archiving
Clients	✓	✓	✓	✓	Full CRUD
Payments	✓	✓	✓	✓	Record, delete, refund, PayMongo
Staff/Users	✓	✓	✓	✓	Create, deactivate, Master Key Transfer
Dispatching	✓	✓	✓	✓	Send job orders, view responses
Inventory	✓	✓	✓	✓	Manage equipment stock
Dishes/Packages	✓	✓	✓	✓	Menu and pricing
Financial Reports	✓	✓	—	—	Revenue analytics
Superadmin Console	✓	✓	✓	—	Settings, DB backup, audit log

Pages: `dashboard.php`, `bookings.php`, `clients.php`, `staff.php`, `financial.php`, `inventory.php`, `dishes.php`, `packages.php`, `recipes.php`, `settings.php`, `superadmin.php`, `archive.php`

Role: `frontdesk`

Day-to-day operations. Cannot manage users or view financial analytics.

Module	C	R	U	D
Bookings	✓	✓	✓	—
Clients	✓	✓	✓	—
Payments	✓	✓	—	—
Dispatching	✓	✓	✓	✓

Pages: [frontdesk/dashboard.php](#), [frontdesk/bookings.php](#), [frontdesk/dispatching.php](#), [frontdesk/costing.php](#)

Role: [staff](#)

Field workers. Isolated to job board, inventory, and event reports.

Module	C	R	U	D
My Job Orders	—	✓	✓	accept/decline —
Inventory Dispatch	✓	✓	✓	—
Event Report	✓	✓	—	—
Leave Requests	✓	✓	—	—
Notifications	—	✓	✓	mark-read —

Pages: [staff/dashboard.php](#), [staff/event_report.php](#)

Core Modules & Process Flows

Module 1: Booking Wizard (End-to-End)

1. Select/create client ([nc_name](#) auto-creates new client record).
2. Choose event date — validated: ≥ 3 days lead time, no conflicts ([UNIQUE KEY idx_unique_event_date](#) + [FOR UPDATE](#) row lock inside transaction).
3. Select package (SET A/B/C at 50/75/100 pax) or custom pax count.
4. Choose dishes — enforces [max_main_dishes](#) and [max_desserts](#) per package; excess dishes trigger per-pax [custom_fee](#) surcharges.
5. Add custom items (lechon, etc.) — flat or per-pax pricing.
6. Enter transport fee, event type, dietary notes.
7. **Rush Booking Check:** If within [rush_threshold_hours](#) (120 hrs), forces 100% payment ([rush_dp_percent](#) = 1.0).
8. Pay via cash/GCash/Maya OR initiate PayMongo checkout session.
9. Transaction: [INSERT INTO bookings](#) → [booking_dishes](#) → [booking_custom_items](#) → [payments](#) — all inside [beginTransaction\(\)/commit\(\)/rollback\(\)](#).
10. Audit log + confirmation email queued.

Module 2: Payments & Online Collections

1. Admin records additional payments against booking (POST `/src/api/payments.php`).
2. System SUMs all payments → updates `amount_paid`, `payment_status` (unpaid/partial/paid).
3. **PayMongo Flow**: Checkout session created → client redirected to hosted page → PayMongo POSTs webhook → `webhooks/paymongo.php` validates HMAC-SHA256 signature → inserts payment with `gateway_reference_id` (UNIQUE constraint = idempotency key prevents duplicate recording).
4. `payment_status.php` polled every few seconds from invoice page (AJAX short-polling) until payment confirms.
5. Balance reminders sent manually via admin or automatically by cron (3 days before event).

Module 3: Staff Dispatching

1. Admin opens dispatching panel → Staff Suggestion Engine runs.
2. Engine filters: `leave_requests` (approved leave) + `job_orders` (already booked same date).
3. Calculates recommended count from `waiter_ratio_wedding/waiter_ratio_birthday` settings.
4. Enforces **1 Head Cook per event** rule at API level.
5. `job_orders` inserted with `status = 'pending'`.
6. Staff receives in-app notification + email.
7. Staff accepts/declines → system checks leave conflict + same-date overlap (prevents double-booking).
8. Admin notified in-app + email.

Module 4: Inventory Dispatch & Return

1. Staff dispatches equipment (POST) — `FOR UPDATE` lock checks `current_stock` before deducting.
2. After event: Staff records returns (PUT). System computes `quantity_out - quantity_in`.
3. Shortfall → **auto-logs breakage** to `booking_breakages` (`ON DUPLICATE KEY UPDATE = idempotent`).
4. `booking.breakage_total` + `total_cost` recalculated atomically in same transaction.
5. If booking was archived, it is **automatically unarchived** to allow new charges.

Module 5: Cancellation Management

1. Admin requests cancellation → `booking_cancellations` created (`status = 'requested'`).
2. Forfeiture = `cancel_forfeiture_percent` (50%) × `total_cost`.
3. Finalized → `booking_status = 'cancelled'`, refund payment inserted (negative amount).
4. Un-cancel restores status + reinstates prior payments.

Module 6: Post-Event Report

1. Assigned staff submits report (`actual_start_time`, `actual_end_time`, `overtime_minutes`, `breakage` items).
2. `overtime_total = overtime_minutes × overtime_rate / 60` calculated and stored.
3. `report_submitted_by` + `report_submitted_at` updated. Admin reviews and manually archives.

Module 7: Archive

1. Admin archives a completed paid booking.
2. Snapshot written to `archived_bookings` with full financial summary.

3. `bookings.is_archived = 1`. Hidden from main list by default.

Module 8: Taste-Testing (Schema-Ready)

Tables `taste_testing`, `taste_test_appointments`, `taste_test_feedback` exist in the database. Supports prospect → appointment → feedback → booking conversion. **No frontend/API wired yet** — schema only.

Advanced/Enterprise Features

Feature	Implementation	Why It's Defense-Ready
PayMongo Webhooks	<code>webhooks/paymongo.php</code> — HMAC-SHA256 signature verification. <code>gateway_reference_id</code> UNIQUE constraint prevents duplicate payment recording.	Real-world payment gateway integration with idempotency.
OTP Account Recovery	3-step: Email → OTP verify (rate-limited via <code>otp_attempts</code> table) → Password reset. OTP expires on timeout.	Brute-force protected with IP + email rate limiting.
Master Key Transfer	Admin transfers "master" role to another admin. Logged as <code>admin_master_transfer</code> in audit log. Single-active-admin enforced.	Business succession management. Full audit trail.
AJAX Short Polling	<code>payment_status.php</code> polled every few seconds from invoice page to detect webhook confirmation.	Real-time UX without WebSockets. Industry standard for payment confirmation.
PDO Transactions	All multi-table writes wrapped in <code>beginTransaction()/commit()/rollBack()</code> .	Prevents partial data corruption. Data integrity guaranteed.
Concurrent Booking Guard	<code>SELECT ... FOR UPDATE</code> inside booking transaction prevents simultaneous same-date bookings.	Solves TOCTOU race condition.
Role-Based Audit Log	Every C/U/D writes to <code>audit_log: user_id, action, entity, old_value</code> (JSON), <code>new_value</code> (JSON), <code>ip_address</code> . 445+ entries in production.	Full forensic trail. Enterprise compliance standard.
Cron Worker	CLI-only (403 if browser access). Tasks: email queue, 3-day reminders, balance alerts, login cleanup. PHPMailer with STARTTLS.	Decoupled background processing.
CSRF Protection	<code>requireCsrft()</code> on every state-changing API. Token in session, validated via <code>X-CSRF-Token</code> header.	Prevents Cross-Site Request Forgery. Applied system-wide.

Feature	Implementation	Why It's Defense-Ready
Rush Booking Engine	Calculates hours to event. If < 120 hrs , forces 100% downpayment. Configurable via settings.	Dynamic business rule engine. Zero hardcoding.

Sections 3–5: Database, Design System, QA Audit

3. Database Architecture & ERD

Data Dictionary

users

Column	Type	Description
id	INT UNSIGNED PK	Auto-increment user ID
name	VARCHAR(100)	Full name
email	VARCHAR(150) UNIQUE	Login email
password	VARCHAR(255)	bcrypt hash
role	ENUM('super_admin','admin','frontdesk','staff')	Access role
phone	VARCHAR(20)	Contact number
is_active	TINYINT(1)	1=active, 0=deactivated
job_class	ENUM('head_cook','cook','waiter','server','helper','any','admin','frontdesk')	Staff specialty
created_at	TIMESTAMP	Account creation time

clients

Column	Type	Description
id	INT UNSIGNED PK	Auto-increment
name	VARCHAR(100)	Client full name
email	VARCHAR(150)	Contact email
messenger_link	VARCHAR(255)	Facebook/Messenger profile

Column	Type	Description
phone	VARCHAR(20)	Mobile number
address	TEXT	Home/delivery address
created_at	TIMESTAMP	Record creation

bookings (Core Table — 36 columns)

Column	Type	Description
id	INT UNSIGNED PK	Booking ID
client_id	INT UNSIGNED FK→clients	Who booked
package_id	INT UNSIGNED FK→packages	Package tier chosen (nullable)
event_type	VARCHAR(50)	Wedding/Debut/Birthday/etc.
event_date	DATE UNIQUE	Event date (enforced unique)
event_time	TIME	Start time
event_location	TEXT	Venue
pax_count	INT UNSIGNED	Total guests
base_pax	INT UNSIGNED	Tier base pax count
extra_pax	INT UNSIGNED	Pax above base
base_price	DECIMAL(10,2)	Package base price
extra_cost	DECIMAL(10,2)	Extra pax cost
transport_fee	DECIMAL(10,2)	Delivery/travel fee
surcharge_total	DECIMAL(10,2)	Sum of custom_item prices
breakage_total	DECIMAL(10,2)	Auto-calculated breakage charges
total_cost	DECIMAL(10,2)	Grand total
amount_paid	DECIMAL(10,2)	Sum of all payments
payment_status	VARCHAR(255)	unpaid/partial/paid
booking_status	VARCHAR(255)	pending/confirmed/completed/cancelled
is_archived	TINYINT(1)	1=archived
invoice_token	VARCHAR(255)	Public access token for invoice
paymongo_checkout_id	VARCHAR(255)	PayMongo session ID (cs_xxxx)
dietary_notes	TEXT	Allergy/dietary restrictions
overtime_minutes	INT	Minutes over scheduled end

Column	Type	Description
<code>overtime_rate</code>	DECIMAL(10,2)	Rate per hour
<code>overtime_total</code>	DECIMAL(10,2)	Computed overtime charge
<code>created_by</code>	INT UNSIGNED FK→users	Staff who created booking
<code>report_submitted_by</code>	INT UNSIGNED FK→users	Staff who submitted event report

payments

Column	Type	Description
<code>id</code>	INT UNSIGNED PK	Payment ID
<code>booking_id</code>	INT UNSIGNED FK→bookings	Associated booking
<code>amount</code>	DECIMAL(10,2)	Amount (negative = refund)
<code>payment_method</code>	ENUM('cash','bank_transfer','gcash','maya','paymongo')	How paid
<code>payment_type</code>	ENUM('payment','refund')	Transaction direction
<code>gateway_reference_id</code>	VARCHAR(255) UNIQUE	PayMongo pay_xxxx ID (idempotency key)
<code>reference_no</code>	VARCHAR(100)	Manual GCash/bank reference
<code>payment_date</code>	DATE	Date of payment
<code>recorded_by</code>	INT UNSIGNED FK→users	Staff who recorded it

packages

Column	Type	Description
<code>id</code>	INT UNSIGNED PK	Package ID
<code>set_name</code>	VARCHAR(100)	SET A / SET B / SET C
<code>pax_count</code>	INT UNSIGNED	Base pax for this tier
<code>price</code>	DECIMAL(10,2)	Flat price at base pax
<code>max_main_dishes</code>	INT UNSIGNED	Allowed main dishes

Column	Type	Description
<code>max_desserts</code>	INT UNSIGNED	Allowed desserts
<code>includes_rice</code>	TINYINT(1)	Whether rice is included

Current packages: SET A (50/75/100 pax: ₱28,500/34,500/41,500), SET B (50/75/100: ₱39,500/45,500/52,500), SET C (50/75/100: ₱41,500/62,500/72,500)

dishes

Column	Type	Description
<code>id</code>	INT UNSIGNED PK	Dish ID
<code>name</code>	VARCHAR(100)	Dish name
<code>category</code>	VARCHAR(30)	Beef/Pork/Chicken/Seafood/Vegetables/Pasta/Rice/Dessert
<code>meal_type</code>	VARCHAR(100)	breakfast/lunch/dinner/all
<code>base_pax</code>	INT	Base pax for this dish
<code>custom_fee</code>	DECIMAL(10,2)	Extra per-pax charge if dish is selected
<code>is_active</code>	TINYINT(1)	Whether dish is bookable

71 dishes across 8 categories.

booking_dishes (Junction)

Links bookings to selected dishes. `UNIQUE KEY uq_booking_dish(booking_id, dish_id)`.

booking_custom_items

Free-text add-ons per booking (lechon, juice, etc.) with category and price.

booking_breakages

Column	Type	Description
<code>booking_id</code>	FK→bookings	Associated booking
<code>equipment_id</code>	FK→equipment	Which item broke
<code>quantity</code>	INT UNSIGNED	Number broken
<code>unit_price</code>	DECIMAL(10,2)	Snapshoted replacement cost
<code>total_cost</code>	DECIMAL(10,2)	qty × unit_price
<code>charge_to</code>	ENUM('client','staff','business')	Who bears the cost
<code>logged_by</code>	FK→users	Who recorded it

`UNIQUE KEY uq_bb_booking_equipment(booking_id, equipment_id)` — prevents duplicate breakage entries for same item.

booking_cancellations

Full financial record of cancellation including `deposit_forfeit`, `refundable_amount`, `refund_method`, `refund_processed_by`.

booking_staff / job_orders

`job_orders` is the active dispatching table. `booking_staff` is the legacy table (empty). `job_orders` tracks: `staff_id`, `role_required`, `status` (pending/accepted/declined), `job_class` (snapshotted at dispatch time).

equipment

Column	Type	Description
<code>name</code>	VARCHAR(100)	Equipment name
<code>category</code>	VARCHAR(50)	General/Dinnerware
<code>current_stock</code>	INT	Available units (deducted on dispatch)
<code>total_stock</code>	INT	Total owned
<code>replacement_cost</code>	DECIMAL(10,2)	Cost if broken

booking_inventory

Tracks each dispatch event: `quantity_out`, `quantity_in`, `dispatched_by`, `dispatched_at`, `returned_by`, `returned_at`. UNIQUE KEY `uq_booking_equipment(booking_id, equipment_id)`.

leave_requests

Staff leave requests with `leave_date`, `status` (pending/approved/rejected), `reviewed_by`. UNIQUE KEY `uq_staff_date(staff_id, leave_date)`.

notifications

Column	Type	Description
<code>recipient_id</code>	INT nullable	Specific user (NULL = role-broadcast)
<code>target_role</code>	VARCHAR(30)	'global' / 'admin' / 'frontdesk'
<code>type</code>	VARCHAR(50)	user_management/booking/finance/dispatch/system
<code>message</code>	TEXT	Human-readable notification body
<code>action_url</code>	VARCHAR(500)	Deep-link URL
<code>is_read</code>	TINYINT(1)	Read state

audit_log

Column	Type	Description
<code>user_id</code>	INT UNSIGNED	Who performed action
<code>action</code>	VARCHAR(60)	e.g. <code>payment_recorded</code> , <code>booking_archived</code>
<code>entity</code>	VARCHAR(30)	<code>booking/payment/client/user/setting</code>
<code>entity_id</code>	INT UNSIGNED	ID of affected record
<code>old_value</code>	LONGTEXT JSON	State before change
<code>new_value</code>	LONGTEXT JSON	State after change
<code>ip_address</code>	VARCHAR(45)	IPv4/IPv6 of actor

`email_queue`

Async email pipeline. `status` ENUM: `pending/sending/sent/failed`. `attempts` max 3. Processed by cron.

`settings`

Key-value store for all runtime configuration. `type` ENUM: `string/int/float/bool/json`. 50+ settings including financial ratios, SMTP credentials, operating hours.

`password_resets`

OTP storage: `email`, `otp` (6-digit INT), `expires_at`. Cleared after successful verification.

`login_attempts`

Brute-force tracking: `ip_address`, `email`, `attempted_at`. Cleaned up by cron after 24h.

`otp_attempts`

Separate rate-limit table for OTP verification: `email`, `ip_address`, `attempted_at`.

`recipe_ingredients`

Ingredients per dish: `dish_id`, `ingredient_name`, `base_quantity`, `unit`, `unit_price`, `supplier`. 513 ingredient records.

`archived_bookings`

Immutable snapshot of completed bookings: `original_id`, client info, financial summary, `archived_by`.

`migrations`

Tracks applied SQL migrations with filename, batch, and SHA-256 checksum.

`taste_testing` / `taste_test_appointments` / `taste_test_feedback`

Schema-ready tables for the taste-testing pipeline (not yet wired to frontend).

[illegible]

Design Framework

- **Color Palette:** System green (`#30D158`) as primary accent — the same green used in iOS. Neutral grays derived from `rgba(60,60,67,x)` opacity scales. Dark mode backgrounds using deep charcoal.
- **Typography:** System font stack (`-apple-system, BlinkMacSystemFont, Segoe UI, Roboto`) for a native OS feel. Font weights span 400–800, used deliberately for hierarchy.
- **Glassmorphism:** Cards use `backdrop-filter: blur()`, semi-transparent backgrounds, and `0.5px` hairline borders (`rgba(60,60,67,0.08)`).
- **Micro-animations:** Sidebar link transitions, table row hover lifts, button press feedback, toast slide-in/out, modal fade effects.
- **Icon Library:** Font Awesome 6 used throughout for semantic icons.
- **Component Library:** Custom card system (`card, card-header, card-body`), `stat-icon` badges (color-coded: teal, sage, amber, coral), `badge-status` pills (paid/partial/unpaid/confirmed/cancelled).

Transactional emails use **inline CSS Apple-style cards**: green gradient headers for bookings, orange gradient for balance reminders, matching the in-app design system.

- **Bootstrap 5 Grid:** `col-lg-X` breakpoints used throughout for two-column to single-column collapsing.
- **Sidebar:** Collapsible hamburger menu on mobile, fixed rail on desktop.

- **Tables:** Horizontal scroll wrappers (`overflow-x: auto`) with sticky first columns where needed.
- **Modals:** Full-screen on mobile (`modal-fullscreen-sm-down`).
- **Staff Dashboard:** Specifically optimized for mobile — large tap targets, card-based layout instead of table, inventory dispatch with mobile-friendly quantity inputs.
- **Pagination:** Custom CSS pagination component in `assets/css/pagination.css` with responsive button sizing.

5. QA Audit: Edge Cases & Known Issues

Edge Cases Successfully Handled

Scenario	How It's Handled
Same-date double booking	<code>UNIQUE KEY idx_unique_event_date(event_date) + SELECT ... FOR UPDATE</code> inside transaction. Two concurrent requests cannot both succeed.
Duplicate PayMongo webhook delivery	<code>UNIQUE KEY uq_gateway_ref(gateway_reference_id)</code> on <code>payments</code> table. Second webhook for same <code>pay_xxxx</code> ID silently fails insert — no duplicate payment.
Inventory over-dispatch	<code>SELECT current_stock ... FOR UPDATE</code> before deducting. Throws exception if <code>current_stock < quantity</code> . Transaction rolled back.
Breakage re-recording	<code>booking_breakages</code> uses <code>ON DUPLICATE KEY UPDATE</code> . Multiple return submissions for same equipment update in place — never create duplicate rows.
Return quantity > dispatched	<code>PUT</code> endpoint throws: "Returned quantity cannot exceed dispatched quantity" and rolls back.
Staff double-booking	Accept endpoint checks <code>job_orders</code> for accepted status on same <code>event_date</code> . Rejects with <code>409 Conflict</code> and conflict details.
Staff on leave + accepting	Accept endpoint queries <code>leave_requests</code> before allowing accept. Rejects with clear message.
Rush booking payment bypass	System checks <code>diffHours < rush_threshold_hours</code> before accepting downpayment — forces 100% for short-notice bookings.
Session expired mid-booking	<code>\$creatorId <= 0</code> check with descriptive message: "Your session has expired. Open a new tab."
PayMongo mock key bypass	<code>paymongo_checkout.php</code> detects <code>sk_test_your_secret_key_here</code> and returns a simulated response — dev testing without real API key.
PDO transaction failures	All multi-step operations wrapped in <code>try/catch(Throwable)</code> with <code>rollback()</code> . Error message surfaced to caller.
Head Cook duplication	API enforces max 1 <code>head_cook</code> per booking with explicit <code>409</code> response.
Booking 1-year max	<code>\$eventDate > date('+1 year')</code> check in booking creation.

Scenario	How It's Handled
Cancelled booking payment	<code>paymongo_checkout.php</code> guard: <code>if (\$booking_status === 'cancelled') → 409</code> . Cannot pay for cancelled booking.
Archive unarchived on breakage	If returning inventory reveals breakage on an archived booking, system auto-unarchives it for financial correction.
OTP brute-force	<code>otp_attempts</code> table tracks failed attempts. <code>checkOtpRateLimit()</code> blocks further attempts.
Login brute-force	<code>login_attempts</code> tracks by IP. Lockout duration configurable (<code>lockout_duration_minutes</code>).
Cron web access	<code>cron_worker.php</code> checks <code>php_sapi_name() !== 'cli'</code> and returns <code>403 CLI only</code> .

Known Vulnerabilities & Bugs

Honest assessment for defense preparation.

Issue	Severity	Details
SMTP password in <code>settings</code> table	 HIGH	<code>smtp_pass</code> stored in plaintext in the <code>settings</code> table (DB). The audit log shows it was even logged in <code>old_value/new_value</code> JSON on update. If DB is compromised, Gmail App Password is exposed. Fix: Encrypt at rest or move to <code>.env</code> only.
<code>audit_log</code> initiated_by null bug	 MEDIUM	Recent conversation (ID: <code>aee33e6f</code>) identified that <code>initiated_log</code> field records NULL in some audit entries. The <code>auditLog()</code> function reads <code>\$_SESSION['user_id']</code> but the session may not be active for cron-triggered operations. Fix: Pass <code>\$userId</code> explicitly to all audit calls.
<code>booking_inventory</code> collation mismatch	 MEDIUM	<code>booking_inventory</code> uses <code>utf8mb4_general_ci</code> while most tables use <code>utf8mb4_unicode_ci</code> . This can cause JOIN performance issues and unexpected collation errors on string comparisons.
<code>max_admins</code> setting orphaned	 MEDIUM	Setting exists in DB (<code>max_admins = 5</code>) but the single-active-admin refactor removed the enforcement logic. The setting is now misleading — it does nothing. Fix: Remove the setting key from DB or add a deprecation note.
No rate limit on manual payment recording	 MEDIUM	<code>payments.php</code> POST has no rate limiting. A fast loop (as demonstrated by <code>audit_log</code> entries 70–89 showing 20+ ₱1 payments in rapid succession) can flood the payments table. Fix: Add a minimum amount validation (e.g., $\geq$ ₱1.00) and optionally a per-booking rate limit.

Issue	Severity	Details
booking_staff table is orphaned	● LOW	booking_staff table exists in schema with 0 rows. job_orders has replaced it. The table wastes schema space and causes confusion. Fix: Drop the table, remove FK references.
Taste-testing tables have no UI	● LOW	3 taste-testing tables exist with full FK relationships but zero frontend. Dead schema. Fix: Either implement or document as "v2 feature."
debug_mode does not fully disable all paths	● LOW	debug_mode setting exists and is checked in some places, but it's not consistently enforced across all API endpoints. Some error messages may still leak stack traces.
No server-side file upload validation	● LOW	No file upload feature exists currently, but if added, max_file_upload_mb setting is defined without corresponding server-side enforcement.
PayMongo amount_pesos logged incorrectly	● LOW	audit_log entry for paymongo_checkout_created logs amount_pesos: balanceDue but the actual amount charged may be the override_amount . The log value may not match the real checkout amount in override scenarios.

Sections 6–9: Security, API, Deployment & Defense Strategy

6. Security Architecture

Authentication & Session Management

Login Flow

1. **POST /src/api/auth.php** — receives **email** + **password**.
2. Checks **login_attempts** for lockout (configurable **max_login_attempts**, **lockout_duration_minutes**).
3. Fetches user row — verifies **is_active = 1** AND **debug_mode = 0** (debug mode blocks all logins).
4. **password_verify()** against bcrypt hash — **no plaintext passwords stored**.
5. On success: session regenerated (**session_regenerate_id(true)**), CSRF token generated (**bin2hex(random_bytes(32))**), **login_attempts** cleared.
6. On failure: **login_attempts** incremented. After threshold, **locked_until** set.

Session Control

- **Session Timeout:** Configurable **session_timeout_minutes** (default: 480 minutes). Enforced on every page load via **requireRole()**.

- **Session Fixation Prevention:** `session_regenerate_id(true)` called on login.
- **CSRF Tokens:** Generated per-session, required in `X-CSRF-Token` header for all POST/PUT/DELETE API calls.
- **Role Enforcement:** `requireRole($role)` checks both `$_SESSION['role']` and session expiry on every protected page. `requireApiRole($roles)` does the same for API endpoints and returns `401` JSON on failure.

Password Recovery (3-Step OTP)

```
Step 1: POST /src/api/forgot_password.php { email }
        → generates 6-digit OTP, inserts into password_resets with
        expires_at
        → queues email via email_queue table

Step 2: POST /src/api/verify_otp.php { email, otp }
        → queries: SELECT id FROM password_resets WHERE email=? AND otp=?
        AND expires_at > NOW()
        → rate-limited via otp_attempts table (checkOtpRateLimit /
        recordOtpAttempt)
        → valid: clears otp_attempts

Step 3: POST /src/api/reset_password.php { email, otp, new_password }
        → re-verifies OTP (prevents step-skip)
        → password_hash($newPassword, PASSWORD_DEFAULT)
        → updates users.password
        → deletes password_resets row
```

Role-Based Access Control (RBAC)

All API endpoints call `requireApiRole(['admin', 'frontdesk'])` (or similar) as the first line of business logic. The middleware:

1. Checks session exists + not expired.
2. Verifies `$_SESSION['role']` is in the allowed array.
3. Returns `401 Unauthorized` (JSON) if either check fails.
4. Returns the user record for use in the handler.

```
// Example: inventory_dispatch.php
requireApiRole(['admin', 'frontdesk', 'staff']);
requireCsrf();
```

Transport & Data Security

Layer	Implementation
HTTPS	Required on production. <code>.env.example</code> mandates <code>BASE_URL=https://yourdomain.com</code> . PayMongo only sends webhooks to HTTPS endpoints.
SQL Injection Prevention	100% parameterized queries via PDO. No raw string interpolation of user input into SQL.
XSS Prevention	<code>htmlspecialchars()</code> applied to user-provided strings before output in HTML contexts. Settings values and audit log entries escaped on render.
CSRF Protection	<code>X-CSRF-Token</code> header required on all state-changing requests. <code>hash_equals()</code> for timing-safe comparison.
Webhook Signature	PayMongo webhooks validated with <code>hash_hmac('sha256', \$rawBody, \$webhookSecret)</code> before processing.
Sensitive <code>.env</code>	<code>.htaccess</code> rule <code>Deny from all</code> in <code>config/</code> folder. <code>.env</code> never committed — <code>.env.example</code> shipped instead.
Password Storage	PHP <code>password_hash()</code> with <code>PASSWORD_DEFAULT</code> (bcrypt, cost factor 12+). Never MD5/SHA1.
OTP Expiry	OTPs have <code>expires_at</code> timestamp. <code>AND expires_at > NOW()</code> in every query.
Database Backup	<code>backup.php</code> uses <code>mysqldump</code> via <code>shell_exec()</code> — access restricted to <code>admin</code> role + CSRF.

.htaccess Security Rules

```
# /src/api/.htaccess
Options -Indexes          # Disable directory listing
Header set X-Content-Type-Options "nosniff"
Header set X-Frame-Options "SAMEORIGIN"
Header set X-XSS-Protection "1; mode=block"
```

7. API Endpoint Reference

Endpoint Inventory (28 API files + 1 webhook)

File	Methods	Auth	Description
<code>auth.php</code>	POST, DELETE	—	Login / Logout
<code>bookings.php</code>	GET, POST, PUT, DELETE	admin, frontend	Full booking CRUD + <code>link_staff</code> action
<code>clients.php</code>	GET, POST, PUT, DELETE	admin, frontend	Client management

File	Methods	Auth	Description
<code>packages.php</code>	GET, POST, PUT, DELETE	admin,frontdesk	Package tier management
<code>dishes.php</code> (implied)	GET	all roles	Dish catalog
<code>recipes.php</code>	GET, POST, PUT, DELETE	admin	Recipe + ingredient management
<code>payments.php</code>	GET, POST, PUT, DELETE	admin,frontdesk	Payment recording + refunds
<code>paymongo_checkout.php</code>	POST	admin,frontdesk OR invoice_token	Create PayMongo checkout session
<code>payment_status.php</code>	GET	— (public)	Poll payment confirmation
<code>webhooks/paymongo.php</code>	POST	HMAC sig	PayMongo event handler
<code>dispatching.php</code>	GET, POST, PUT, DELETE	admin,frontdesk,staff	Job order management
<code>inventory.php</code>	GET, POST, PUT, DELETE	admin,frontdesk	Equipment catalog management
<code>inventory_dispatch.php</code>	GET, POST, PUT	admin,frontdesk,staff	Dispatch + return equipment
<code>breakages.php</code>	GET, POST, PUT, DELETE	admin,frontdesk,staff	Breakage log management
<code>staff.php</code>	GET, POST, PUT, DELETE	admin	User/staff account management
<code>settings.php</code>	GET, PUT	admin	Read/update settings
<code>audit_logs.php</code>	GET	admin	Paginated audit log
<code>notifications.php</code>	GET, POST, PUT	all roles	In-app notifications
<code>availability.php</code>	GET	admin,frontdesk	Check event date availability
<code>cancellations.php</code>	GET, POST, PUT, DELETE	admin,frontdesk	Cancellation workflow
<code>event_reports.php</code>	GET, POST, PUT	admin,frontdesk,staff	Post-event reports
<code>leave.php</code>	GET, POST, PUT, DELETE	all roles	Staff leave requests

File	Methods	Auth	Description
analytics.php	GET	admin	Dashboard analytics + charts
archive.php	GET, POST, PUT	admin	Archive management
forgot_password.php	POST	—	OTP generation
verify_otp.php	POST	—	OTP validation
reset_password.php	POST	—	Password reset
send_invoice.php	POST	admin,frontdesk	Queue invoice email to client
backup.php	GET	admin	Trigger mysqldump download

Standard JSON Response Envelope

Every endpoint returns the same envelope:

```
{
  "success": true,
  "message": "Human-readable message",
  "data": { ... }
}
```

HTTP status codes used correctly: 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 409 Conflict, 422 Unprocessable Entity, 500 Internal Server Error, 502 Bad Gateway.

Example: Booking Creation Request (POST [/src/api/bookings.php](#))

```
{
  "client_id": 12,
  "event_date": "2026-06-15",
  "event_time": "10:00",
  "event_location": "Tagaytay Reception Hall",
  "event_type": "Wedding",
  "pax_count": 120,
  "package_id": 3,
  "selected_dishes": [14, 22, 31, 45, 67],
  "transport_fee": 500.00,
  "downpayment": 15000.00,
  "downpayment_method": "gcash",
  "downpayment_ref": "GC-20260615-001",
  "dietary_notes": "2 guests are lactose intolerant",
}
```

```
"notes": "VIP table of 10 at the front"
}
```

Example: PayMongo Webhook Payload (Verified via HMAC-SHA256)

```
{
  "data": {
    "attributes": {
      "type": "payment.paid",
      "data": {
        "attributes": {
          "amount": 1500000,
          "status": "paid",
          "metadata": { "booking_id": 89 }
        },
        "id": "pay_XXXXXXXXXXXXXXXXXXXX"
      }
    }
  }
}
```

8. Deployment Checklist

Pre-Launch Requirements

Environment Configuration (`.env`)

```
# Application
APP_NAME=Yazzies Catering OMS
APP_ENV=production
BASE_URL=https://yourdomain.com/

# Database
DB_HOST=localhost
DB_NAME=yazzie_catering
DB_USER=db_user_here
DB_PASS=strong_password_here

# PayMongo
PAYMONGO_SECRET_KEY=sk_live_XXXXXXXXXXXX
PAYMONGO_PUBLIC_KEY=pk_live_XXXXXXXXXXXX
PAYMONGO_WEBHOOK_SECRET_KEY=whsec_XXXXXXXXXXXX

# SMTP (e.g. Gmail App Password)
MAIL_FROM=noreply@yourdomain.com
MAIL_FROM_NAME=Yazzies Catering
SMTP_HOST=smtp.gmail.com
```

```
SMTP_PORT=587
SMTP_USER=your_gmail@gmail.com
SMTP_PASS=your_app_password_here
SMTP_SECURE=tls
```

Server Requirements

Requirement	Minimum	Recommended
PHP	7.4	8.1+
MySQL / MariaDB	5.7	8.0
Extensions	PDO, PDO_MySQL, mbstring, curl, openssl	+ zip, opcache
RAM	256MB	512MB
Storage	1GB	5GB
SSL	Required	Let's Encrypt (free)

Cron Job Setup (Production cPanel / SSH)

```
# Run every 15 minutes
*/15 * * * * /usr/bin/php /home/username/public_html/cron_worker.php >>
/home/username/logs/cron.log 2>&1
```

PayMongo Webhook Configuration

1. Log into PayMongo Dashboard → Developers → Webhooks.
2. Add URL: <https://yourdomain.com/src/api/webhooks/paymongo.php>
3. Select events: [payment.paid](#), [checkout_session.payment.paid](#).
4. Copy the Webhook Secret → paste into `.env` as `PAYMONGO_WEBHOOK_SECRET_KEY`.

Post-Deployment Verification

- ☐ <https://yourdomain.com/> loads login page correctly.
- ☐ Admin can log in. CSRF token present in `<meta name="csrf-token">`.
- ☐ Create test booking → payment recorded → audit log entry created.
- ☐ PayMongo mock checkout session returns URL (test key in `.env`).
- ☐ Staff can log in → see job board → accept job.
- ☐ Email queue → cron runs → test email received.
- ☐ `cron_worker.php` returns 403 when accessed from browser.
- ☐ `backup.php` downloads valid `.sql` file.
- ☐ OTP password reset flow completes end-to-end.

9. System Defense Strategy

Anticipated Examiner Questions & Model Answers

Q1: "Why did you choose PHP over a modern framework like Laravel or Node.js?"

A: The decision was deliberate and client-driven. PHP runs on any shared hosting (₱2,500/year) without a Node.js server or Docker. For a catering SME operating in a Philippine province, zero-configuration deployment is a business requirement, not a technical compromise. The codebase follows MVC-adjacent separation: `config/` (infrastructure), `includes/` (middleware), `src/api/` (controllers), `views/` (templates). The custom router in `index.php/.htaccess` handles routing. This demonstrates that good architecture is a design decision — not a framework dependency.

Q2: "Why is the database not normalized to 3NF? I see `client_name` repeated in `archived_bookings`."

A: `archived_bookings` is an **intentional denormalized snapshot**. When a booking is archived 6 months later, the client's name may have changed. Financial and legal records require immutable historical snapshots. This is the same design pattern used in invoice systems worldwide — the invoice stores the client's name at the time of issue, not a foreign key that points to a mutable record. This is a deliberate trade-off between normalization and audit fidelity.

Q3: "Is the PayMongo integration production-ready?"

A: Yes. The integration implements all four pillars of production-grade payment processing: (1) **Idempotency** — `gateway_reference_id` UNIQUE constraint prevents duplicate charges from re-delivered webhooks. (2) **Signature Verification** — HMAC-SHA256 validation of every webhook payload. (3) **Balance Validation** — system re-reads balance inside a DB transaction to prevent stale reads. (4) **Guard Rails** — cancelled bookings and fully paid bookings are blocked from generating new checkout sessions with descriptive `409 Conflict` responses. The developer mock bypass (`sk_test_your_secret_key_here`) allows full UI testing without live API credentials.

Q4: "What happens if two staff members book the same event date simultaneously?"

A: The booking creation API uses a `SELECT COUNT(*) FROM bookings WHERE event_date = :date FOR UPDATE` inside a PDO transaction. The `FOR UPDATE` clause acquires a row-level lock in InnoDB, serializing concurrent requests for the same date. The first transaction to acquire the lock succeeds; the second finds `count > 0` and returns a `409 Conflict` with "This date was just booked by someone else." The database also has a `UNIQUE KEY idx_unique_event_date(event_date)` as a final backstop.

Q5: "How do you prevent a staff member from being assigned to two events on the same date?"

A: The job acceptance endpoint (`PUT /dispatching.php`) runs two queries before updating: (1) `SELECT FROM leave_requests WHERE staff_id=? AND leave_date=? AND status='approved'` — blocks acceptance if on approved leave. (2) `SELECT FROM job_orders jo JOIN bookings b ... WHERE jo.staff_id=? AND jo.status='accepted' AND b.event_date=? AND jo.booking_id != thisBid` — blocks acceptance if already committed to another event on the same date. Both queries return `409 Conflict` with the conflicting record IDs for full traceability.

Q6: "How does the system handle broken equipment charges?"

A: When staff record inventory returns via `PUT /inventory_dispatch.php`, the system computes `diff = quantity_out - quantity_in`. If `diff > 0`, a breakage record is inserted/updated in `booking_breakages` (idempotent via `ON DUPLICATE KEY UPDATE`). The `bookings.breakage_total` and `bookings.total_cost` are then **recalculated atomically** in the same transaction using a subquery `SUM`, and `payment_status` is re-evaluated (paid/partial/unpaid) based on the new total. All of this happens in one database transaction — either all succeeds or all rolls back.

Q7: "What are the most critical security vulnerabilities?"

A: (Honest, pre-prepared answer shows academic integrity.) Three identified risks: (1) **SMTP password in DB** — `smtp_pass` is stored in the `settings` table. Mitigated by DB access controls, but ideally should be in `.env` only. (2) **Audit log session dependency** — cron-triggered operations can log NULL `user_id`. Fix: pass explicit system user ID. (3) **Debug mode** — if accidentally left `enabled`, all logins are blocked system-wide. Mitigated by a UI warning label in the Superadmin Console.

Q8: "What is the Superadmin Console for?"

A: It is the system's infrastructure management panel, restricted to the `admin` role. It provides: (1) **Global Configuration** — live-editable settings (SMTP, security thresholds, pricing rules) organized into Security / Email / System tabs. Changes take effect immediately across all modules via the `settings` table cache. (2) **Database Backup** — one-click `mysqldump` download of a full SQL snapshot. (3) **System Activity Log** — paginated audit trail of all actions, auto-refreshes on setting change.

Q9: "How would you scale this if Yazzies grew from 5 bookings/month to 500?"

A: Current stack handles 500 bookings/month easily. For scale: (1) **Read replicas** — offload `SELECT` queries from the `analytics.php` and financial reports to a MySQL read replica. (2) **Redis cache** — cache `settings` table (queried on every page load) and `dishes` catalog. (3) **Queue worker** — replace `email_queue` + cron polling with a proper queue (Beanstalkd, Redis Queue) for sub-second email delivery. (4) **CDN** — offload `assets/` (CSS, JS, images) to Cloudflare or AWS CloudFront. (5) **Object storage** — move database backups from server filesystem to S3. None of these require rewriting business logic — the API layer is already stateless and scalable.

Q10: "What is the `invoice_token` for?"

A: Each booking is issued a cryptographically random 32-character hex token (`bin2hex(random_bytes(16))`) stored in `bookings.invoice_token`. This token enables **passwordless, client-facing invoice access**. The invoice URL is: `https://domain.com/templates/invoice.php?booking_id=89&token=abc123...`. The `paymongo_checkout.php` endpoint accepts this token as an alternative to a staff session, enabling the client to pay directly from the invoice link on their phone — without creating an account. The token is validated via constant-time comparison to prevent timing attacks.

Defense Presentation Outline (30-min oral defense)

Time	Topic	Key Points
0:00–3:00	System Demo	Show login → create booking → dispatch staff → record payment → view audit log

Time	Topic	Key Points
3:00–8:00	Problem & Solution	Present Section 1. Emphasize the "no localhost" argument for production deployment.
8:00–15:00	Architecture Deep-Dive	ERD walkthrough. bookings as central entity. Explain UNIQUE constraint + FOR UPDATE.
15:00–20:00	Security Features	CSRF, bcrypt, OTP flow, PayMongo HMAC, role middleware. Show a real audit log entry.
20:00–25:00	Q&A Defense	Use the model answers above. Be honest about the smtp_pass issue — show you identified it.
25:00–30:00	Roadmap	Mention: v2.0 PDF invoices, SMS integration, Redis caching, taste-testing module UI completion.

Proof-of-Enterprise Talking Points (Lead with These)

1. **Idempotent payment processing** — industry-standard pattern. "PayMongo can send the same webhook 3 times. Our UNIQUE constraint prevents triple-charging the client."

2. **TOCTOU Race Condition fix** — "We solved the concurrent booking problem using database-level locking, not application-level checks."

3. **PDO transaction rollback** — "If inserting the dishes fails after inserting the booking, the booking is also rolled back. Partial data is impossible."

4. **Cron Worker as infrastructure** — "We designed background processing as a separate concern. The **cron_worker.php** is CLI-only — it cannot be triggered from the web."

5. **Role-based audit log** — "Every financial mutation has a paper trail. We can reconstruct the exact state of any booking at any point in time from the audit log."

6. **Master Key Transfer** — "Business succession is built into the system. The ownership of the admin account can be transferred formally, with a logged record." ENDOFFILE

End of Report. Sections: 6 (Security), 7 (API Reference), 8 (Deployment), 9 (Defense Strategy) Generated from live codebase analysis — 28 API files, 24 DB tables, 513 recipe ingredients, 445+ audit log entries.